Taming Multi-Dimensional Tail: A Factorized Tail-Interaction Framework for Hyperscale Cloud Workload Forecasting

Xin Wu[*]
Southwest Jiaotong University
Chengdu, China
xinwu@my.swjtu.edu.cn

Fei Teng[*] [†]
Southwest Jiaotong University
Chengdu, China
Engineering Research Center of Sustainable Urban Intelligent Transportation
Chengdu, China
fteng@swjtu.edu.cn

Chen Yang[*]
Southwest Jiaotong University
Chengdu, China
yangc@swjtu.edu.cn

Tianrui Li[*]
Southwest Jiaotong University
Chengdu, China
trli@swjtu.edu.cn

Abstract

The adoption of AI-native cloud infrastructure is reshaping workload behavior in large-scale data centers, moving from stationary, homogeneous patterns to highly dynamic, heterogeneous ones. An empirical analysis of production cloud monitoring data shows that system reliability is increasingly governed by non-linear interactions in the multi-dimensional tail space, including long-range temporal backlogs, spatial resource fragmentation, and irreversible state rigidity. Accurately forecasting these tail-dominated behaviors is essential for capacity planning and reliability management. However, existing forecasting models face two limitations: learning is dominated by abundant normal workloads, leading to rare but critical tail events being underrepresented, and reliance on linear composition assumptions limits the ability to model joint tail anomalies across correlated temporal, spatial, and state dimensions. To address these challenges, we introduce the Factorized Tail-Interaction Framework (FTIF), a data-driven forecasting and risk perception framework explicitly designed for tail-sensitive cloud workloads. Firstly, an adaptive trend decoupler separates dominant workload trends from tail residuals using a gradient detachment mechanism and learnable gating mechanisms. Secondly, a factorized resonance module augments attention mechanisms with hyper-order factorization, injecting multiplicative inductive biases to capture non-linear interactions among multi-dimensional risk factors. Thirdly, a rank-rectified tail-focal optimization employs topology-preserving reweighting to align the learning objective with rare but operationally critical tail events. Experiments on real-world cloud datasets show that FTIF achieves an MSE of 0.0015 on the GenTD26 dataset and captures 94.95% of tail events in Spot scenarios, enabling finer-grained risk perception and robust modeling of joint tail anomalies.

CCS Concepts

• **Networks** → **Cloud computing**; • **Applied computing** → **Forecasting**.

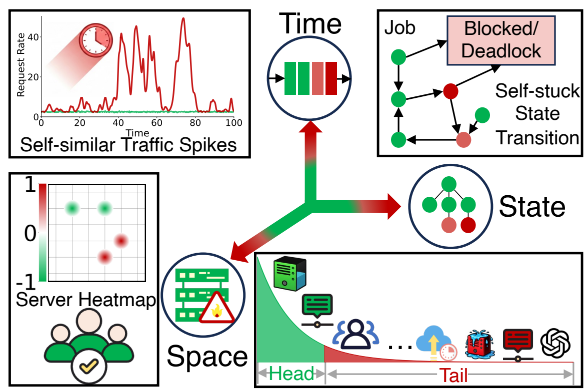

Figure 1: Illustration of multi-dimensional long-tail effects across time, state, and space in cloud systems.

Keywords

Cloud Workload Forecasting, Multi-Dimensional Tail Risk

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1 Introduction

The evolution of cloud infrastructure toward AI-native architectures has fundamentally altered workload characteristics, shifting from stationary stochastic processes to highly dynamic, non-linear regimes [18, 21]. Unlike traditional microservices, large-scale GenAI tasks exhibit resource rigidity, burst volatility, and long-range dependencies [24, 30]. These attributes increase system complexity, rendering workload forecasting a challenge that transcends univariate time series regression and requires reconstructing high-dimensional system states to prevent failures [14]. Consequently, workload forecasting has evolved beyond a simple univariate time

[*]School of Computing and Artificial Intelligence, Southwest Jiaotong University
[†]Fei Teng is the corresponding author.

series regression problem into a complex system-engineering challenge that requires high-dimensional state reconstruction. The objective transcends mere resource scheduling efficiency and becomes a critical foundation for preventing system-level failures.

This transformation is particularly critical in hyperscale data centers deploying cloud-native [3], microservices [25], and serverless computing [8] architectures. In these environments, SLOs for mission-critical applications are rarely constrained by average load but are instead dominated by rare long-tail events occurring beyond the P95 or P99 quantiles [9, 19, 22]. Furthermore, production reliability boundaries are not defined by scalar thresholds in a single dimension but are governed by a complex multi-dimensional tail phenomenon. As illustrated in Fig. 1, this risk stems from the non-linear coupling of sparse tail features across distinct physical dimensions. Specifically, burst requests in the temporal dimension trigger non-Markovian queue backlogs that result in latency tails, while multi-tenant resource contention in the spatial dimension leads to pareto-style resource polarization and severe memory fragmentation. Simultaneously, long-cycle job retention in the state dimension forces the system into highly rigid and irreversible states. This structural resonance across time, space, and state implies that system failures often occur at specific coupling moments when the resource space is highly fragmented and rigid, even if the aggregate load remains well below peak capacity.

Recent years have seen notable progress in cloud workload forecasting. Transformer-based models such as DynEformer [11] leverage global receptive fields to capture long-range temporal dependencies, while MetaEformer [12] introduces meta-learning to adapt to evolving system dynamics. State-space-model-based approaches, including SDE-inspired architectures [29], further enhance continuous-time modeling capabilities. Despite their success in fitting the bulk distribution, these methods consistently underperform in the extreme tail regime, where accurate prediction is most critical for system robustness. We argue that this failure is not incidental, but structural. Existing models face two fundamental challenges when applied to long-tail cloud workloads. **Firstly, normality-dominated gradient bias.** In large-scale, highly imbalanced data streams, the overwhelming majority of samples correspond to normal operating conditions. During gradient-based optimization, these abundant normal samples generate dense and coherent gradient flows that dominate parameter updates. In contrast, gradients induced by rare extreme events are sparse and often treated as noise. This optimization bias causes models to converge toward stable manifolds of high-probability states, effectively suppressing representations that are crucial for long-tail risk modeling. **Secondly, joint tail distortion under additive inductive bias.** Most existing architectures rely on linear combination or additive feature fusion, implicitly assuming that system risk accumulates additively across dimensions. However, extreme cloud failures often arise from non-linear multiplicative synergies across time, resource, and system states—for example, when high concurrency coincides with memory fragmentation and scheduling contention. Such joint tail behavior cannot be faithfully approximated by additive models, leading to systematic distortion of joint tail distributions and underestimation of coupled extreme risks.

These observations reveal a deeper issue: long-tail prediction failure stems from a mismatch between learning dynamics, representation structure, and the topology of system risk. Addressing this mismatch requires models to satisfy three necessary conditions: (i) explicitly decouple tail dynamics from dominant normal trends to prevent gradient suppression; (ii) possess sufficient structural capacity to model high-order non-linear interactions across dimensions; (iii) align the optimization objective with the ordinal structure of system risk rather than pointwise error alone.

To this end, we propose the Factorized Tail-Interaction Framework (FTIF), a principled modeling paradigm designed specifically for long-tail coupled risks in non-stationary cloud systems. FTIF establishes a dedicated high-fidelity computational pathway for tail dynamics, independent of the dominant bulk distribution. To mitigate gradient dominance, FTIF introduces a Tail-Sensitive Embedding (TSE) that amplifies extreme signals via intensity-aware injection and quantile-based gating, followed by an Adaptive Trend Decoupler (ATD). ATD, together with a Trend Detachment Operator (TDO), explicitly enforces a structural gradient barrier. This mechanism isolates the residual optimization path, forcing tail representations to be learned in a subspace protected from interference from the dominant trend gradient. To address joint tail distortion, we design a Factorized Resonance Module (FRM) that embeds high-order factorization operators into the attention mechanism's value stream. By introducing explicit multiplicative inductive bias, FRM enables accurate reconstruction of non-linear synergistic effects among multiple risk factors, overcoming the expressive limitations of additive architectures. Finally, to ensure stable tail learning under extreme imbalance, we propose the Rank-Rectified Tail-Focal Optimization (RTO) strategy. Rather than treating prediction as independent pointwise regression, RTO deploys a risk-aware loss that incorporates quantile-aware ranking constraints and dynamic rank rewards, correcting the topology of empirical risk minimization and aligning model predictions with the true ordinal structure of system risk. The main contributions of this work are threefold:

- We formalize the multi-dimensional tail coupling in hyperscale workloads and empirically identify the representation collapse of additive models under state rigidity and gradient dominance.
- We propose FTIF, an end-to-end paradigm that decouples dominant normal trends from tail dynamics, introduces multiplicative inductive bias to model non-linear risk coupling, and aligns learning dynamics with the true topology of long-tail system risk under extreme imbalance.
- Extensive experiments on large-scale real-world cloud production datasets demonstrate that FTIF consistently outperforms 16 baselines, achieving superior global accuracy and significantly improved sensitivity to P95 extreme events.

2 Preliminary

2.1 Problem Definition

We consider the problem of cloud workload forecasting within a hyperscale non-stationary environment. Let $\mathcal{X} = \{\mathbf{x}_t\}_{t=1}^{T}$ denote a

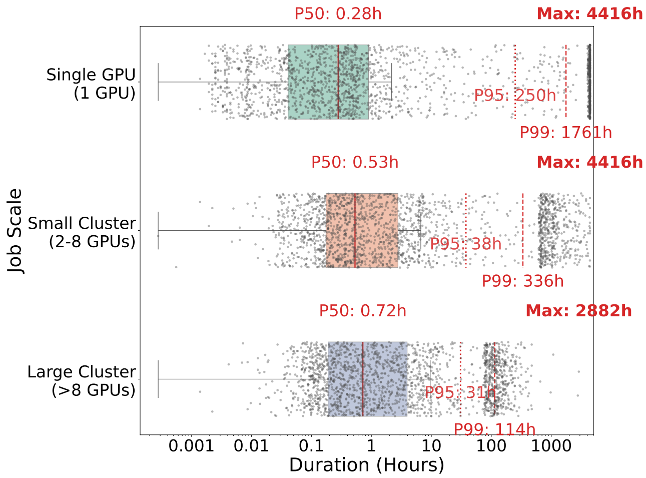

Figure 2: Evidence of irreversible state rigidity.

continuous sequence of system state vectors, where $\mathbf{x}_t \in \mathbb{R}^D$ represents a D-dimensional feature vector comprising server load, resource utilization, and job status at time step t. Given a historical observation window of length L, denoted as $\mathbf{X}_t = [\mathbf{x}_{t-L+1}, \cdots, \mathbf{x}_t] \in \mathbb{R}^{L\times D}$, the goal is to accurately predict the future states for the subsequent H time steps, denoted as $\mathbf{Y}_t = [\mathbf{x}_{t+1}, \cdots, \mathbf{x}_{t+H}] \in \mathbb{R}^{H\times D}$. Mathematically, we aim to learn a mapping function $\mathcal{F}_\theta : \mathbb{R}^{L\times D} \to \mathbb{R}^{H\times D}$ parameterized by θ that minimizes the discrepancy between the predicted sequence $\hat{\mathbf{Y}}_t$ and the ground truth $\mathbf{Y}_t$.

Unlike standard forecasting tasks that prioritize the mean accuracy on the dominant normal distribution D_{norm}, our primary objective is to capture the dynamics of extreme events located in the sparse long-tail region. We define the tail sample set:

$$S_{tail} = \left\{(t, d) | x_{t,d} > q_\tau\right\} \tag{1}$$

where $\mathbf{x}_{t,d}$ denotes the value of the d-th dimension at time step t. S_{tail} represents the core extreme event set that directly relates to system reliability boundaries. For subsequent optimization alignment, we will extend this set to a broader critical risk set S_{risk} (detailed in §3.4) to cover pre-failure accumulation phases, with $S_{tail} \subset S_{risk}$.

A critical challenge in this context is the non-linear coupling of risks across dimensions. We posit that system failures result from the multiplicative interaction of multi-dimensional factors rather than isolated univariate outliers. Consequently, the problem necessitates modeling the conditional joint probability $P(\mathbf{Y}_t \in \mathcal{S}_{tail} \mid \mathbf{X}_t)$ where dependencies are non-additive. The objective is to minimize a risk-aware loss function $\mathcal{L}_{risk}$ that prioritizes both the magnitude fidelity of $\hat{\mathbf{Y}}_t$ in $\mathcal{S}_{tail}$ and the correct ranking of extreme events.

2.2 Data Analysis

Our empirical analysis of the GenTD26 [33] and Spot26 [6] datasets exposes a fundamental misalignment between the elasticity assumptions of standard forecasting models and the plastic rigidity observed in production environments. While classical queuing theory posits that system states rapidly revert to the mean following load release, cloud systems exhibit multi-dimensional tail coupling.

2.2.1 Irreversible state rigidity. As illustrated in Fig. 2, the distribution of job durations reveals a profound bifurcation between normal and tail regimes. In the single GPU trace, a stochastic disconnect exists between the median (P50) duration of 0.28 hours and the tail (P99) duration of 1,761 hours. This order-of-magnitude decoupling indicates that the system state is not homogeneous but splits into a fluid normal state and a frozen tail state. The persistence of these extreme resident tasks creates a resource lock-in effect that impedes system recovery during congestion. Consequently, standard models optimizing for mean squared error inevitably converge to the dense median region, treating critical tail rigidities as statistical noise and systematically underestimating risk duration.

2.2.2 Temporal and spatial constraints. This state rigidity is further compounded by non-Markovian temporal hysteresis and spatial resource polarization. Latency responses are decoupled from instantaneous load, while resource availability is fragmented by dominant workloads. We provide a rigorous characterization of these coupled phenomena in Appendix B.2 and Fig.9.

3 Methodology

This section elaborates on the FTIF. Addressing the systematic statistical gradient masking of rare signals and the representational incapacity of linear inductive biases to model multiplicative tail interactions in cloud workload forecasting, FTIF does not seek to rectify traditional additive models. Instead, it constructs a high-fidelity extreme-value computation path within a unified differentiable architecture, independent of the normal distribution. As illustrated in Fig. 3, the computational flow comprises three tightly coupled stages. Firstly, the TSE (§3.1) explicitly injects extreme-value energy via nonlinear magnitude mapping in the input manifold space, preventing the dissipation of sparse risk signals in the shallow network layers. Subsequently, the ATD (§3.2) employs an orthogonal gradient shielding operator to construct a topologically isolated computational subspace. This mechanism physically severs the gradient interference of dominant normal trends on long-tail residuals. Finally, the FRM (§3.3) introduces a hyper-order factorization to precisely reconstruct the multiplicative synergy of multi-dimensional risk factors within the latent space. The entire learning process is supervised by the RTO (§3.4), which uses a dynamic rank-reward mechanism to align the model's optimization trajectory with the true risk topology.

3.1 Tail-Sensitive Embedding

Standard linear embedding layers, particularly when combined with input normalization (e.g., Z-score), tend to squash high-magnitude outliers towards a zero-mean distribution. This process inherently obscures the distinct statistical properties of extreme events, leading to a loss of magnitude information. To address this limitation, the TSE is designed to explicitly preserve the absolute intensity of input signals via a dual-path injection mechanism.

3.1.1 Magnitude-Aware Injection. Drawing inspiration from extreme value theory [16, 26], we posit that the scalar magnitude of a feature vector serves as a critical indicator for distinguishing tail regimes from the bulk distribution. Given an input sequence $\mathcal{X} \in \mathbb{R}^{B\times L\times D_{in}}$, we first obtain a semantic representation $\mathbf{e}_{base}$ via

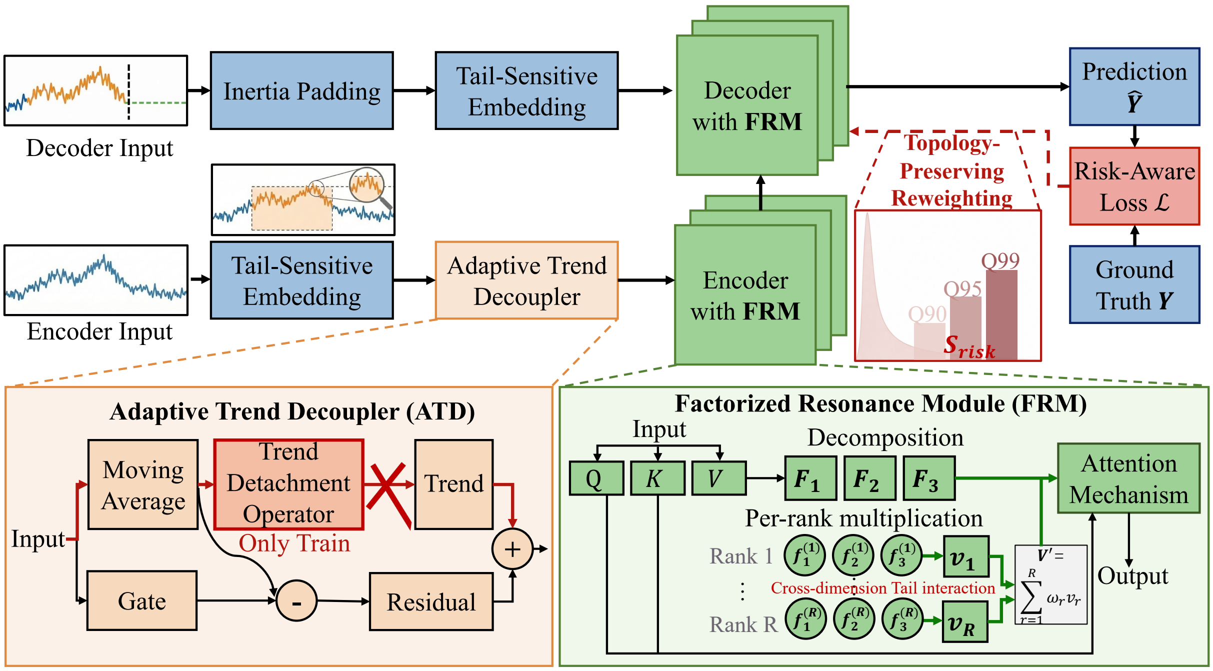

Figure 3: The overall architecture of the FTIF, enabling decoupled trend learning and high-order risk resonance.

linear projection: $\mathbf{e}_{base} = \text{LayerNorm}(\mathbf{x}_t \mathbf{W}_{base})$. To recover lost intensity information, we simultaneously extract the ℓ_2-norm as a scalar intensity metric and project it into the high-dimensional latent space:

$$\mathbf{e}_{mag} = \text{Tanh}(\text{MLP}_{enh}(\|\mathbf{x}_t\|_2)) \in \mathbb{R}^{D_{model}}. \tag{2}$$

Here, MLP_{enh} transforms the scalar norm into a magnitude embedding vector, while the Tanh activation limits the amplitude to prevent numerical instability during deep forward propagation.

3.1.2 Risk-Gated Modulation. To enforce differentiated processing for high-risk samples, we introduce a conditional modulation mechanism. The risk identification relies on a quantile threshold q_τ. To ensure robust inference without batch dependence, we compute dynamically across the batch during training to adapt to local variations, and use a fixed global moving-average threshold derived from training statistics during inference.

The binary risk mask is defined as $\mathbb{M}_{tail} = \mathbb{I}(\|\tilde{\mathbf{x}}_t\|_2 > q_\tau)$. Crucially, to rigorously prevent look-ahead bias (data leakage), $\tilde{\mathbf{x}}_t$ denotes the effective input token initialized via Inertia Padding (detailed in Appendix C.2) for future time steps, strictly independent of the ground truth. The final embedding $\mathbf{e}_t$ is synthesized as:

$$\mathbf{e}_t = \mathbf{e}_{base} + (1 + \lambda \cdot \mathbb{M}_{tail}) \odot \mathbf{e}_{mag} \odot \sigma(\text{MLP}_{gate}(\tilde{\mathbf{x}}_t)), \tag{3}$$

where λ is a learnable scalar gain and $\odot$ denotes element-wise multiplication. This formulation acts as a risk-gated bias: when a sample is identified as a tail event ($\mathbb{M}_{tail} = 1$), the contribution of $\mathbf{e}_{mag}$ is explicitly amplified, ensuring the representation is robustly augmented by intensity features.

3.2 Adaptive Trend Decoupler

The ATD is designed to mitigate the *gradient washout* phenomenon, where dominant low-frequency trend components overwhelm the optimization landscape. Unlike standard decomposition schemes that allow joint optimization, the ATD enforces a structural constraint to physically isolate the residual subspace from the trend manifold.

3.2.1 Trend-Residual Decomposition. We extract the low-frequency trend $\mathbf{T}_{raw} = \text{AvgPool}(\mathbf{H}_{in})$ from the latent representation $\mathbf{H}_{in}$. To prevent the trend reconstruction from dominating the learning process, we employ a TDO, denoted as $\Psi(\cdot)$, to generate a gradient-detached trend $\mathbf{T}_{clean} = \Psi(\mathbf{T}_{raw})$. The residual component $\mathbf{H}_{res}$, which primarily encapsulates high-frequency tail signals, is then derived as:

$$\mathbf{H}_{res} = \text{Proj}_{res}(\mathbf{H}_{in} - \mathbf{g} \odot \mathbf{T}_{clean}), \tag{4}$$

where $\mathbf{g} = \sigma(\text{MLP}(\mathbf{H}_{in}))$ is the adaptive modulation coefficient regulating the intensity of trend removal.

3.2.2 Optimization Dynamics via Residual Gradient Flow. The TDO $\Psi(\mathbf{x})$ acts as a stop-gradient barrier that functions as an identity mapping during forward propagation but blocks backward flow:

$$\text{Forward: } \Psi(\mathbf{x}) = \mathbf{x}, \quad \text{Backward: } \frac{\partial \Psi(\mathbf{x})}{\partial \mathbf{x}} \equiv \mathbf{0}. \tag{5}$$

While this blocks the direct feedback from the trend component it does not leave the encoder untrained. Instead, it reconfigures the total gradient flow to the encoder θ_{enc} to be driven exclusively through the residual path:

$$\frac{\partial \mathcal{L}}{\partial \theta_{enc}} = \frac{\partial \mathcal{L}}{\partial \mathbf{H}_{res}} \cdot \underbrace{\left(\frac{\partial \mathbf{H}_{res}}{\partial \mathbf{H}_{in}} \cdot \frac{\partial \mathbf{H}_{in}}{\partial \theta_{enc}} \right)}_{\text{Residual Gradient Path}} . \tag{6}$$

Since $\mathbf{H}_{res}$ is obtained by subtracting $\mathbf{T}_{clean}$ from $\mathbf{H}_{in}$, this mechanism effectively performs a *gradient subtraction*. By zeroing out the trend-related gradient components that typically possess much larger magnitudes, we force the encoder to prioritize the representation of sparse, high-frequency perturbations. This prevents the encoder from collapsing into a trivial low-pass filter and ensures that the optimization process remains sensitive to tail risk signals.

3.3 Factorized Resonance Module

To address the challenge where the linear superposition assumption $\mathbf{O} = \text{Attn}(\mathbf{Q}, \mathbf{K})\mathbf{V}$ inherent in standard attention mechanisms fails to capture the multiplicative synergy of multi-dimensional risks (e.g., the intersection of high concurrency and resource fragmentation), the FRM introduces an explicit multiplicative inductive bias via a Hyper-Order Factorization mechanism applied to the Value stream.

3.3.1 Hyper-Order Factorization. Let $\mathbf{V} \in \mathbb{R}^{B \times L \times D}$ be the value tensor derived from the input representation. We project $\mathbf{V}$ into M latent factor subspaces to disentangle independent risk components. For the m-th factor ($m \in [1, M]$), we compute the factor projection:

$$\mathbf{F}_m = \mathbf{V}\mathbf{W}^{(m)}, \quad \text{where } \mathbf{W}^{(m)} \in \mathbb{R}^{D \times R}. \tag{7}$$

Here, R represents the factorization rank, effectively compressing the feature space into compact risk manifolds.

3.3.2 Rank-wise Resonance and Reconstruction. To capture the non-linear synergy among distinct factors, we execute a rank-wise resonance operation. Unlike linear aggregation, we compute the element-wise Hadamard product across factors at each rank index $r \in [1, R]$ to model the conjunction of risks:

$$\mathbf{v}_r = \mathbf{F}_{1,:,r} \odot \mathbf{F}_{2,:,r} \odot \cdots \odot \mathbf{F}_{M,:,r} \in \mathbb{R}^{B \times L}. \tag{8}$$

This operation ensures that a resonance term $\mathbf{v}_r$ is activated if and only if all M risk factors exhibit high magnitude simultaneously at rank r. The reconstructed value tensor $\mathbf{V}' \in \mathbb{R}^{B \times L \times D}$ is then obtained by aggregating these resonance terms via a learnable projection $\mathbf{W}_{out} \in \mathbb{R}^{R \times D}$:

$$\mathbf{V}' = \text{Concat}(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_R)\mathbf{W}_{out}. \tag{9}$$

3.3.3 Physical Interpretation. The FRM fundamentally differs from standard attention by modulating the content ($\mathbf{V}$) rather than the routing ($\mathbf{Q}, \mathbf{K}$). While $\mathbf{Q}\mathbf{K}^\top$ determines where to attend, our resonance mechanism determines what information is retrieved based on multiplicative validity. By injecting this high-order synergistic information into the global attention field ($\mathbf{H}_{out}$), FTIF reconstructs the structural resonance mechanism of system failures, ensuring that tail risks are not merely summed but amplified through factor coincidence.

3.4 Rank-Rectified Tail-Focal Optimization

Standard regression objectives are inherently biased towards the majority of normal samples, often treating rare high-value events as noise. To mitigate this gradient dominance problem, we propose the RTO, which integrates multi-criteria risk identification with a batch-stabilized topology constraint.

3.4.1 Multi-Criteria Risk Identification. Relying solely on extreme statistical thresholds yields sparse supervision, as it fails to capture the gradual accumulation of risk factors preceding extreme events. We thus define a broader critical risk set S_{risk} that acts as a buffer zone encompassing S_{tail} and high-load transition samples:

$$\mathcal{S}_{risk} = \underbrace{\{i | y_i > Q_\tau\}}_{S_{tail}\ (\tau=0.95)} \cup \{i | y_i > \mu_y + 1.5\sigma_y\} \cup \{i | y_i \in TopK_{15\%}(y)\} .$$

Here, Q_τ is the same quantile threshold as in S_{tail} to ensure consistency. By incorporating the top 15% of high-load samples and samples exceeding the mean plus 1.5 times the standard deviation, the model sensitizes itself to the pre-failure accumulation phase while retaining focus on core extreme events in S_{tail}.

3.4.2 Topology-Preserving Reweighting. To handle the high variance of cloud workloads, we decompose the sample weight into severity-aware modulation and rank-topology constraints.

Severity-Aware Modulation. We employ a dynamic focal weight to amplify the gradient response for samples with substantial prediction errors. The base weight ω_i for sample i is formulated as:

$$\omega_i = 1 + \alpha \left(\frac{|y_i - \hat{y}_i|}{\max(|\mathbf{y} - \hat{\mathbf{y}}|) + \epsilon} \right)^\gamma + \lambda \cdot \mathbb{I}(i \in \mathcal{S}_{risk}), \tag{10}$$

where γ is the focal parameter and λ is the tail boost factor ensuring that risk samples receive sufficient gradient attention.

Rank-Consistency Rectification. To prevent mean-reversion, where extreme values are predicted as averages, we introduce a rank-consistency coefficient η_i. Let $\text{Perc}(\hat{y}_i)$ be the percentile rank of the prediction within the current batch. The rectification term is defined as:

$$\eta_i = \begin{cases} 1 - \beta, & \text{if } i \in \mathcal{S}_{risk} \text{ and } \text{Perc}(\hat{y}_i) \geq \tau_{rank} \\ 1 + \delta, & \text{if } i \in \mathcal{S}_{risk} \text{ and } \text{Perc}(\hat{y}_i) < \tau_{rank} \\ 1, & \text{otherwise} \end{cases} , \tag{11}$$

where $\tau_{rank} = \tau - 0.1$ is a relaxed rank threshold. This mechanism provides a rank-reward (lowering β) for correctly identified risks and an additional rank-penalty (increasing δ) for missed risks that are predicted in low-rank percentiles, thereby forcing the model to preserve the relative topological ordering of extreme events.

3.4.3 Composite Risk Objective. The final objective function $\mathcal{L}$ synthesizes the reweighted regression loss with a tail-specific relative error constraint to optimize both magnitude and percentage accuracy:

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^{B} (\eta_i \cdot \omega_i \cdot \mathcal{L}_{Huber}(y_i, \hat{y}_i)) + \xi \cdot \text{MAPE}_{\mathcal{S}_{risk}}(\mathbf{y}, \hat{\mathbf{y}}). \tag{12}$$

The conditional $\text{MAPE}_{\mathcal{S}_{risk}}$ term explicitly minimizes the relative percentage error within the critical risk zone, ensuring operational reliability for capacity planning.

Table 1: Performance comparison on the two datasets (mean ± std). Lower values indicate better performance for metrics (MSE, MAE, and Tail-MAPE@95), while higher values are preferred for Tail-CR@95. Best and second-best results are highlighted in blue and light blue, respectively. Improvement denotes the relative gain of FTIF over the best competing baseline.

Method	GenTD26				Spot26			
	MSE	MAE	Tail-MAPE@95 (%)	Tail-CR@95	MSE	MAE	Tail-MAPE@95 (%)	Tail-CR@95
PatchTST	0.1137±0.0107	0.2465±0.0135	147.86±95.19	0.2834±0.0213	0.0210±0.0002	0.1128±0.0005	113.27±2.83	0.4882±0.0037
TimeXer	0.1247±0.0097	0.2440±0.0087	376.09±25.93	0.2967±0.0499	0.0202±0.0007	0.1104±0.0015	110.65±2.22	0.4900±0.0089
SDE	0.0156±0.0001	0.0840±0.0001	83.42±0.25	0.2427±0.0014	0.0706±0.0002	0.2112±0.0009	51.69±0.76	0.4060±0.0016
MoU	0.1479±0.0267	0.2768±0.0197	63.95±4.99	0.0339±0.0672	0.0137±0.0001	0.0857±0.0002	69.18±0.10	0.2726±0.0013
Chronos-2	0.1322±0.0111	0.2654±0.0090	163.03±29.49	0.2484±0.0447	0.0214±0.0003	0.1136±0.0006	113.20±5.57	0.4826±0.0075
iTransformer	0.1585±0.0099	0.2975±0.0120	574.49±53.18	0.3717±0.0281	0.0211±0.0004	0.1127±0.0010	108.93±0.84	0.4935±0.0085
Crossformer	0.0912±0.0051	0.2414±0.0073	172.42±61.34	0.2184±0.0109	0.0194±0.0002	0.1072±0.0005	114.30±2.24	0.4859±0.0063
TSMixer	0.1197±0.0082	0.2768±0.0088	144.64±96.41	0.2100±0.0149	0.0218±0.0002	0.1156±0.0005	119.76±1.68	0.4350±0.0081
FusAD	0.1900±0.0389	0.3277±0.0383	36.50±5.93	0.3691±0.0432	0.0364±0.0003	0.1559±0.0006	152.22±0.94	0.3148±0.0083
TiDE	0.1778±0.0397	0.3076±0.0283	139.14±85.63	0.2017±0.0038	0.0268±0.0005	0.1287±0.0013	134.49±1.32	0.3762±0.0123
AutoXPCR	0.2311±0.1939	0.2856±0.1195	12.23±0.40	0.5800±0.0473	0.0033±0.0001	0.0152±0.0002	136.87±6.24	0.5145±0.0050
DLinear	0.1186±0.0038	0.2506±0.0081	304.01±52.30	0.2084±0.0060	0.0227±0.0001	0.1174±0.0002	124.23±0.94	0.4636±0.0028
MetaEFormer	0.6450±0.1315	0.5689±0.0659	8.01±1.30	0.6200±0.0000	0.0251±0.0008	0.1247±0.0024	39.42±1.02	0.3872±0.0259
TimeFilter	0.1222±0.0090	0.2535±0.0113	297.46±53.39	0.2934±0.0389	0.0208±0.0002	0.1123±0.0006	112.64±2.40	0.4885±0.0069
DynEformer	0.2274±0.0294	0.4051±0.0208	10.8300±0.14	0.4525±0.0794	0.0131±0.0000	0.0859±0.0004	40.44±0.05	0.7586±0.0108
OrgLinear	0.0127±0.0004	0.0872±0.0018	24.50±1.07	0.5591±0.0464	0.0137±0.0001	0.0846±0.0000	75.97±0.10	0.6382±0.0008
FTIF (Ours)	0.0015±0.0003	0.0299±0.0022	2.06±0.51	0.7426±0.0879	0.0001±0.0000	0.0006±0.0001	33.37±5.44	0.9495±0.0048

4 Experiments

In this section, we empirically evaluate FTIF's effectiveness in capturing coupled multi-dimensional tail risks and forecasting extreme transients within hyperscale cloud workloads. Specifically, our experiments aim to answer the following research questions:

- **RQ1:** How does FTIF compare against state-of-the-art baselines in terms of global forecasting accuracy and, more critically, the capability to capture extreme tail risks across diverse cloud workload scenarios?
- **RQ2:** What is the individual and synergistic contribution of each proposed component, particularly the ATD, FRM, and RTO, to the final forecasting performance?
- **RQ3:** How does FTIF behave under varying hyper-parameter configurations and severe environmental degradations?
- **RQ4:** Can FTIF provide physically meaningful insights into latent risk factors and maintain robust performance under real-world system degradations?

4.1 Experiment Settings

4.1.1 Datasets. We conducte a systematic evaluation of the proposed framework on two large-scale, industrial-grade trace datasets released from Alibaba clusters. These datasets capture key frontier challenges in GenAI services and heterogeneous GPU resource management, respectively. The GenTD26 dataset [33] characterizes the end-to-end execution behavior of stable diffusion inference pipelines in production environments. We aggregate GPU memory utilization at a 60s temporal granularity, enabling fine-grained analysis of resource dynamics and temporal fluctuations across multi-stage GenAI workloads. This dataset provides a unique, realistic experimental foundation for studying the time-varying resource demands inherent to complex GenAI inference pipelines. The Spot26 dataset [6] comprises execution traces from 4,278 heterogeneous GPU nodes, including A100 and A10 GPUs, and comprehensively records the co-execution behavior of high-priority jobs and opportunistic jobs in shared-resource settings. On Spot26 dataset, we focus on job duration prediction as the primary task. This target exhibits a pronounced long-tailed distribution, while the relationships between job-level metadata and node hardware configurations are highly complex and non-linear. Detailed descriptions are provided in Appendix B.

4.1.2 Baselines. To ensure a comprehensive and rigorous evaluation, we benchmark FTIF against 16 state-of-the-art baselines spanning four representative modeling paradigms. **Firstly,** for general forecasting models, we include PatchTST [17], TimeXer [28], SDE [29], and MoU [20] as exemplars of deep representation learning, together with Chronos-2 [1] as a recent foundation model. This comparison assesses whether FTIF exhibits superior sensitivity to tail risks relative to conventional MSE-driven representations, regardless of whether they are instantiated via Transformer-based or state-space architectures. **Secondly,** within interaction models, we compare against iTransformer [15], Crossformer [36], and TSMixer [4], aiming to expose the limitations of static interaction mechanisms and to demonstrate the necessity of synergistic tensor gating for capturing conditional failure resonances. **Thirdly,** for decomposition-based models, we evaluate FusAD [35], TiDE [5], AutoXPCR [7], and DLinear [34]. **Finally,** we consider domain-specific models, including MetaEformer [12], TimeFilter [10], DynEformer [11], and OrgLinear [6], to assess the robustness of FTIF in modeling multi-dimensional tail risks inherent in cloud physics, beyond the capabilities of specialized vertical solutions.

4.1.3 Evaluation metrics. We evaluate forecasting performance using a combination of global accuracy and tail-sensitive metrics. Mean Squared Error (MSE) and Mean Absolute Error (MAE) measure overall predictive accuracy. To assess performance in extreme

Table 2: Ablation study of different components in FTIF.

Index	Component			GenTD26				Spot26			
	ATD	FRM	RTO	MSE	MAE	Tail-MAPE@95 (%)	Tail-CR@95	MSE	MAE	Tail-MAPE@95 (%)	Tail-CR@95
A	✓	✗	✗	0.0024±0.0006	0.0368±0.0045	5.06±1.45	0.6516±0.0818	0.0168±0.0005	0.0107±0.0002	42.11±6.41	0.9203±0.0103
B	✗	✓	✗	0.0024±0.0008	0.0365±0.0055	3.48±1.61	0.7253±0.0813	0.0168±0.0009	0.0107±0.0001	41.51±5.81	0.9389±0.0093
C	✗	✗	✓	0.0023±0.0008	0.0379±0.0065	4.57±1.94	0.7023±0.0924	0.0124±0.0009	0.0145±0.0001	38.12±4.02	0.9396±0.0079
D	✓	✓	✗	0.0018±0.0004	0.0360±0.0024	4.31±1.07	0.6469±0.0612	0.0046±0.0002	0.0017±0.0002	39.34±6.16	0.9431±0.0097
E	✗	✓	✓	0.0021±0.0005	0.0379±0.0040	3.98±0.56	0.6285±0.0227	0.0031±0.0001	0.0019±0.0001	35.87±3.39	0.9462±0.0057
F	✓	✗	✓	0.0016±0.0002	0.0399±0.0017	2.81±1.08	0.6687±0.0644	0.0011±0.0002	0.0015±0.0001	33.46±3.19	0.9495±0.0048
FTIF (Ours)	✓	✓	✓	**0.0015±0.0003**	**0.0299±0.0022**	**2.06±0.51**	**0.7426±0.0879**	**0.0001±0.0000**	**0.0006±0.0001**	**33.37±5.44**	**0.9495±0.0048**

regimes, we report Tail-MAPE@95, which computes relative errors on samples above the 95th percentile, and Tail Capture Rate (Tail-CR@95), which quantifies the proportion of true tail events correctly captured by the model.

4.2 Overall Performance (RQ1)

Table 1 presents a systematic comparison between FTIF and 16 baselines across the GenTD26 and Spot26 datasets. The empirical results demonstrate that FTIF establishes a new performance envelope across both global forecasting accuracy and tail risk capture. **Firstly,** in terms of MSE and MAE, FTIF exhibits robust adaptability to non-stationary workload dynamics. On the GenTD26 dataset, FTIF achieved an MSE of 0.0015, representing an order-of-magnitude improvement over the second-best baseline, OrgLinear (0.0127). Similarly, on the Spot26 dataset characterized by high-frequency transient volatility, FTIF reduced the error floor to 0.0001 and significantly outperformed the strongest decomposition-based baseline, MoU (0.0137). Unlike PatchTST and iTransformer, which tend to generate over-smoothed predictions due to the dominance of low-frequency trends, the ATD explicitly isolates gradient interference, allowing the model to preserve high-fidelity signals. **Secondly,** FTIF demonstrates a fundamental advantage in capturing extreme regimes. Standard architectures, constrained by additive superposition assumptions, exhibit substantial deviations in the long-tail region; for instance, on GenTD26, Crossformer and DLinear recorded Tail-MAPE@95 values of 172.42% and 304.01%, respectively. In contrast, FTIF constrained this tail deviation to a remarkable 2.06%. **Finally,** regarding the recall of critical failures, FTIF achieved a Tail-CR@95 of 0.9495 on the Spot26 dataset. Compared with the competitive DynEformer model, the proposed model achieved only 0.7586, implying a miss rate of approximately 24% for critical tail events. This improvement highlights the RTO's critical role. By dynamically reweighting the loss landscape based on sample difficulty, RTO mitigates the gradient masking phenomenon caused by extreme data imbalance, ensuring that sparse failure signals are not suppressed by the dense normal distribution.

4.3 Ablation Study (RQ2)

Table 2 validates the indispensability of each module within FTIF. The complete framework consistently outperforms all ablated indexes (A-F). **Firstly,** comparing Index E (w/o ATD) with the full model reveals that removing the ATD results in a catastrophic 30-fold increase in MSE (0.0001 to 0.0031) on Spot26. **Secondly,** the FRM is essential for capturing coupled risks. Excluding FRM (Index F) degrades Tail-MAPE@95 on GenTD26 from 2.06% to 2.81%,

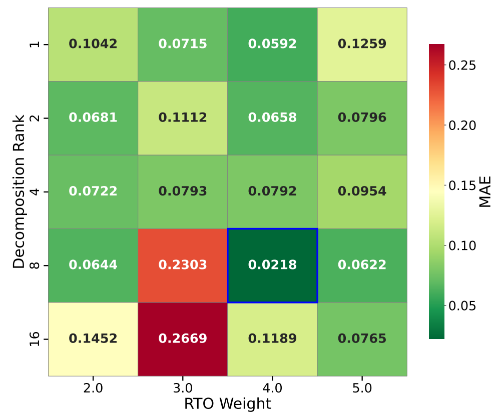

Figure 4: Parameter sensitivity analysis of decomposition rank and RTO weight.

indicating that standard additive aggregation fails to approximate joint tail distributions. The multiplicative inductive bias introduced by FRM effectively models the non-linear resonance between risk factors. **Finally,** the RTO module (Index D vs. Full) improves Tail-CR@95 coverage from 94.31% to 94.95%. By re-weighting the optimization landscape based on sample difficulty, RTO prevents the model from collapsing into safe, mean-value predictions, ensuring sensitivity to rare events.

4.4 Robustness and Reliability Analysis (RQ3)

4.4.1 Hyperparameter Sensitivity. The sensitivity analysis on GenAI workloads reveals that the robustness of FTIF hinges on the synergy between representation capacity (decomposition rank) and constraint intensity (RTO weight). As shown in Fig. 4, performance improves consistently as the rank increases from 1 to 8, achieving a global optimum at rank 8 with an MAE of 0.0218. This confirms that a moderate tensor rank effectively expands the hypothesis space of the FRM, enabling the reconstruction of complex nonlinear synergies without underfitting. However, excessive capacity introduces optimization risks. At rank 16, insufficient constraint intensity leads to gradient instability, causing the MAE to surge to 0.2669. In this regime, the RTO weight acts as a critical stabilizer. Increasing the weight to 4.0 or 5.0 enhances tail signals and suppresses high-dimensional oscillations, restoring the MAE to a robust range. These results underscore an intrinsic mechanism: high-capacity models require stronger risk alignment constraints

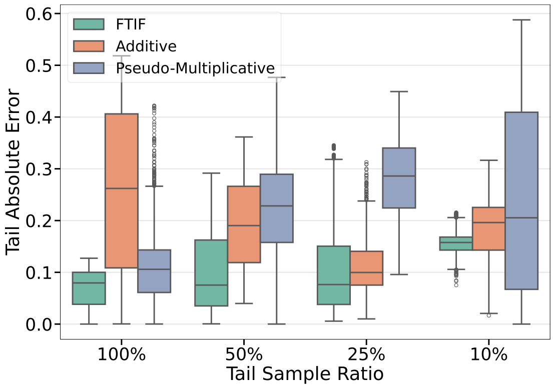

Figure 5: Robustness evaluation under tail data sparsity, showing FTIF's superior resilience compared to additive and pseudo-multiplicative baselines.

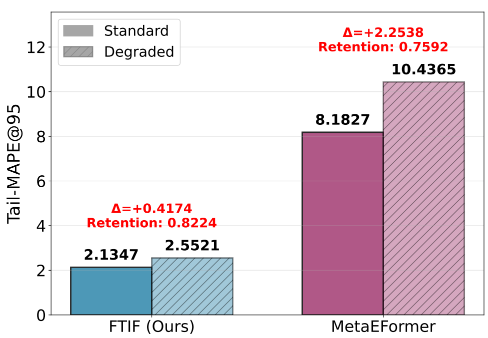

Figure 6: Stress-testing results under composite degradation scenarios.

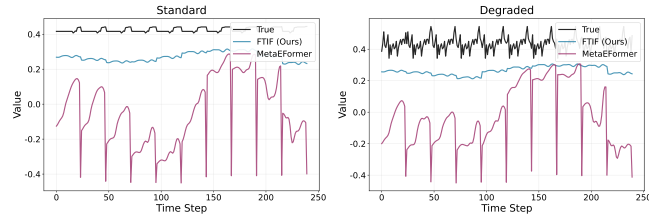

Figure 7: Visualization of prediction error distributions.

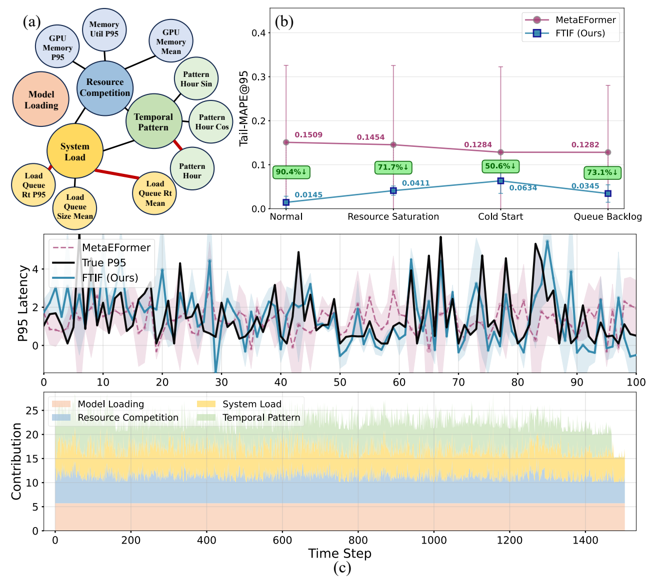

Figure 8: Case study of FTIF interpretability and robustness in P95 latency modeling.

to maintain optimization stability in the presence of non-stationary dynamics.

4.4.2 Structural Resilience to Tail Sparsity. As shown in Fig. 5, we evaluate the robustness of FTIF under data sparsity by systematically reducing the availability of tail samples to 100%, 50%, 25%, and 10% in the GenTD26. Even when tail sample availability drops to 10%, FTIF demonstrates superior structural inference capability. Its median tail absolute error remains consistently below 0.15 with a compact distribution, showing no signs of divergence. In contrast, the additive interaction exhibits persistent underfitting, with errors exceeding 0.2 across all sample abundances, constrained by its linear assumptions. Under sparse data distributions, the pseudo-multiplicative interaction suffers from severe generalization failure. Its error variance explodes to over 0.6 at the 10% ratio, which indicates that unconstrained point-wise Hadamard products fail to infer unseen combinatorial risks when critical co-occurrence samples are missing. Conversely, FTIF leverages the low-rank factorization within the FRM to decompose high-order interactions into independently learnable latent factors. This mechanism enables the precise reconstruction of multi-dimensional coupled risks even on manifolds characterized by extreme data scarcity.

4.4.3 Stress-Testing under Composite Degradation. We empirically verify predictive resilience under composite environmental deterioration via composite degradation settings that include tail sparsification, noise injection, and rank constraints. As illustrated in Fig. 6, quantitative stress tests reveal that structural stress significantly impacts the baseline MetaEFormer. Its Tail-MAPE@95 deteriorates notably, resulting in a retention rate of 0.7592, indicating high sensitivity to distribution shifts. In contrast, FTIF leverages the structural robustness of the Factorized Resonance Mechanism to suppress non-stationary noise interference. This effectively limits the tail error increment to a low level and maintains a superior retention rate of 0.8224. The temporal visualization in Fig. 7 further confirms this stability. Even on degraded manifolds with missing features and reduced signal-to-noise ratios, FTIF maintains smooth, bounded prediction trajectories and effectively avoids the severe oscillations observed in the baseline model.

4.5 Case Study (RQ4)

We evaluate the interpretability and robustness of FTIF in decomposing P95 latency within GenAI workloads. Fig. 8 provides a consolidated case study. Fig. 8 (a) illustrates an interpretable taxonomy of latent risk factors, including resource competition, system load, and temporal patterns. Fig. 8 (b) quantifies model performance across four distinct system states. The results show that FTIF consistently minimizes Tail-MAPE@95 compared with the MetaEFormer baseline, reducing prediction error by 90.4% in the Normal phase and retaining substantial advantages under volatile cold-start and queue-backlog conditions. This demonstrates FTIF's ability to withstand structural stress and distribution shifts.

Fig. 8 (c) further corroborates this robustness through temporal visualization. FTIF accurately tracks high-frequency fluctuations in true P95 latency, avoiding the pronounced lag and underestimation observed with the baseline. Moreover, the dimension contribution panel dynamically instantiates the conceptual categories shown in Fig. 8 (a) into the time domain. By disentangling aggregate latency into independent semantic trajectories, FTIF confirms that its predictive gains arise from precise isolation of multi-source interference rather than memorization of surface-level patterns.

5 Conclusion

In conclusion, this paper formulates cloud workload forecasting as a multi-dimensional long-tail modeling problem and shows that prediction failures primarily stem from the dominance of normal regimes and additive interaction assumptions. We propose FTIF, which isolates tail dynamics via trend decoupling, captures nonlinear coupling through factorized interactions, and aligns learning with tail risk under extreme imbalance. Extensive experiments on real-world cloud workloads demonstrate consistent gains in both global accuracy and tail-event capture.

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A Algorithmic Details and Complexity Analysis

A.1 Algorithm Details

The training process for our FTIF is shown in Algorithm 1. The implementation, configurations, and preprocessing scripts for FTIF are available at https://github.com/wuliwuxin/FTIF.

A.2 Complexity Analysis

Despite incorporating high-order tensor interactions and topological decoupling mechanisms, FTIF rigorously maintains a computational efficiency profile comparable to the standard Transformer through low-rank approximation and operator optimization.

A.2.1 Time Complexity. Given an input sequence of length L and feature dimension D, the computational bottleneck of the standard Transformer lies in the self-attention mechanism, with a complexity of $O(L^2D + LD^2)$. The additional overhead in FTIF primarily stems from three lightweight modules.

The TSE and ATD mainly involve element-wise operations and shallow MLP projections. Their combined complexity is $O(L \cdot D^2)$, which scales linearly with sequence length and is negligible compared to the quadratic cost of the attention mechanism.

In the FRM, we reconstruct the standard value projection (with $O(L \cdot D^2)$ complexity) into a tensor factorization form. Specifically, the complexity for manifold factorization and rank-wise reconstruction is $O(M \cdot L \cdot D \cdot R)$. Since we employ a low-rank constraint in our design (where $R \ll D$, e.g., $R = 32, D = 512$) and the interaction order M is a small constant, the actual computational load of FRM is significantly lower than that of full-rank linear projection. Consequently, the total time complexity of FTIF is $O(L^2D + L \cdot D \cdot R)$, which remains strictly within the $O(L^2D)$ order of magnitude.

A.2.2 Parameter Efficiency. Benefiting from the compressive nature of factorization, FRM reduces the parameter count of the feature interaction layer from $O(D^2)$ to $O(M \cdot D \cdot R)$. When processing high-dimensional data, this parameter-sharing mechanism not only reduces the memory footprint but also serves as an implicit regularization, effectively mitigating the risk of overfitting to sparse, long-tail samples.

Algorithm 1: Training Procedure of FTIF

Input: Dataset $\mathcal{D}$; Learning rate η; Batch size B; Rank R; Tail Quantile τ.
Output: Optimized parameters Θ^*.

Initialize model parameters Θ randomly.
while *not converged* **do**
  Sample mini-batch $\mathcal{B} = \{(\mathbf{X}, \mathbf{Y})\} \sim \mathcal{D}$;
  // Phase 1: Tail-Sensitive Embedding (§3.1)
  Compute magnitude energy: $\mathbf{m} \leftarrow \|\mathbf{X}\|_2$;
  Determine dynamic threshold: $q_\tau \leftarrow \text{Quantile}(\mathbf{m}, \tau)$;
  Generate tail mask: $\mathbb{M}_{\text{tail}} \leftarrow \mathbb{I}(\mathbf{m} > q_\tau)$;
  $\mathbf{E} \leftarrow \texttt{TSE}(\mathbf{X}, \mathbf{m}, \mathbb{M}_{\text{tail}})$;
  // Phase 2: Adaptive Trend Decoupler (§3.2)
  $\mathbf{T}_{\text{raw}} \leftarrow \text{AvgPool}(\mathbf{E})$;
  $\mathbf{H}_{res} \leftarrow \text{Proj}_{res}\left(\mathbf{E} - \texttt{TDO}(\mathbf{T}_{\text{raw}}) \odot \text{Gate}(\mathbf{E})\right)$;
  // Phase 3: Factorized Resonance Module (§3.3)
  for $l \leftarrow 1$ **to** L **do**
    Project subspaces: $\mathbf{Q}, \mathbf{K}, \mathbf{V} \leftarrow \text{Proj}_l(\mathbf{H})$;
    Map $\mathbf{V}$ to factor subspaces: $\{\mathbf{F}^{(1)}, \cdots, \mathbf{F}^{(K)}\} \leftarrow \texttt{Decomp}(\mathbf{V})$;
    Aggregate resonance: $\mathbf{V}' \leftarrow \sum_{r=1}^{R} \omega_r \bigodot_{k=1}^{K} \mathbf{F}^{(k)}_{:,r}$;
    Update hidden states: $\mathbf{H} \leftarrow \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}') + \mathbf{H}$;
  end
  // Phase 4: Rank-Rectified Tail-Focal Optimization (§3.4)
  Forecast: $\hat{\mathbf{Y}} \leftarrow \text{Decoder}(\mathbf{H})$;
  Identify tail set $\mathcal{S}_{\text{tail}}$ via prediction error and magnitude;
  Compute Loss: $\mathcal{L} \leftarrow \sum_{i \in \mathcal{S}_{\text{tail}}} \omega_i \ell(\hat{y}_i, y_i) + \lambda \mathcal{L}_{\text{rank}}$;
  Update parameters: $\Theta \leftarrow \Theta - \eta \nabla_\Theta \mathcal{L}$.
end

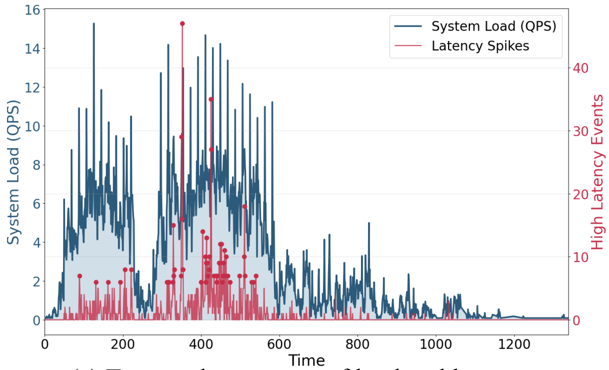
A.2.3 Theoretical Conclusion. Unlike baseline models that introduce complex recurrence or convolutions, FTIF achieves Pareto optimality between high-order expressiveness and computational economy via algebraic dimensionality reduction rather than simple module stacking.

B Theoretical Foundations and Data Analysis

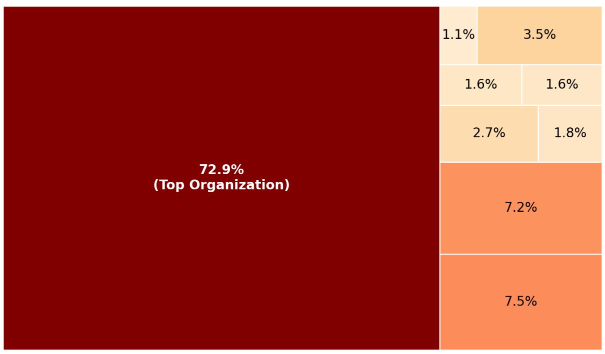
B.1 Formal Definition of State Rigidity

This section details the temporal and spatial mechanisms that exacerbate the state rigidity described in Sec. 2.2.

Definition B.1 (State Rigidity). Given a system mapping $\Phi : \mathcal{X} \to \mathcal{S}$, the region $\mathcal{S}_{rigid} \subset \mathcal{S}$ is defined by the sensitivity collapse: $\|\nabla\Phi(\mathbf{x})\| < \epsilon$ for $\mathbf{x} \in \mathcal{X}_{tail}$. In this regime, standard gradient-based learning often ignores tail signals because they make negligible contributions to the global loss.



(a) Temporal resonance of load and latency.



(b) Spatial skewness showing resource monopoly.

Figure 9: Multi-dimensional tail characteristics: temporal hysteresis and spatial polarization.

B.2 Empirical Data Analysis

B.2.1 Temporal dimension: non-Markovian hysteresis. The relationship between system load and service latency exhibits severe Phase Misalignment, as shown in Fig. 9 (a). High-latency events do not synchronize with load peaks but exhibit significant hysteresis. Quantitative analysis reveals that 45.9% of latency violations occur within the 30-minute window after the workload subsides to normal levels. This establishes that tail risks exhibit a non-Markovian property: the probability of SLA violation depends on the integral of historical backlog rather than the instantaneous arrival rate. This behavior invalidates the stationarity assumption of linear autoregressive models, which fail to capture these deferred repercussions.

B.2.2 Spatial dimension: resource polarization. In the spatial dimension, resource consumption follows a strict Pareto distribution, leading to Resource Polarization. As depicted in Fig. 9 (b), the top 1% of dominant workloads monopolize 72.9% of the total cluster GPU memory. This concentration results in Spatial Fragmentation: while global capacity appears sufficient on average, local allocatable slots are exhausted by a minority of tasks. Consequently, minor fluctuations in the resource usage of these dominant workloads trigger cascading allocation failures for the remaining 99% of tasks. This structural fragility necessitates forecasting architectures capable of modeling high-order interactions between spatial constraints and temporal demands.

C Experimental Design

C.1 Preprocessing

To maintain numerical stability while rigorously preserving critical long-tail signals within highly skewed industrial distributions, we implement a two-stage processing strategy.

C.1.1 Tail-protection normalization. Standard normalization often suppresses extreme tail samples. To counter this, for series exhibiting skewness greater than 1.5, we first apply a logarithmic transformation $\log(1+x)$, followed by a robust scaler based on the median and interquartile range. This pipeline effectively retains the magnitude distinctiveness of tail events.

C.1.2 Tail-aware feature. We further augment the input space with explicit long-tail indicators, including high-order quantile features (P90, P95, P99) and distribution shape metrics. This enhancement enables the model to explicitly encode the boundary characteristics of workload dynamics into the embedding space.

C.2 Inertia Padding

A primary concern in Transformer-based forecasting is the potential for look-ahead bias (data leakage). We address this through a rigorous evaluation protocol and a zero-leakage decoder design. The GenTD26 and Spot26 datasets are strictly partitioned chronologically into training, validation, and testing sets following a 7:1:2 ratio to prevent temporal leakage. We set the look-back window length to $L_{seq} = 96$ and the prediction horizon to $L_{pred} = 24$.

To ensure the isolation of future ground truth from the model input, we construct the decoder input $X_{dec} \in \mathbb{R}^{(L_{token}+L_{pred})\times D}$ using a specialized inertia padding mechanism. The decoder input comprises two segments: a context segment containing the last $L_{token} = 48$ observed steps, and a prediction placeholder. Crucially, strictly forbidding the use of future ground truth or standard interpolation that might imply future trends, we initialize the prediction placeholder by repeating the *last observed state vector* x_t. Mathematically, for the future horizon $k \in \{1, \cdots, L_{pred}\}$, the decoder input is defined as:

$$X_{dec}^{t+k} = x_t, \quad \text{s.t.} \quad X_{dec}^{t+k} \neq x_{t+k}^{groundtruth} \tag{13}$$

This initialization acts as a zero-order hold, ensuring that the ground truth $Y = x_{t+1:t+L_{pred}}$ remains exclusively a supervision target for loss computation, thereby mathematically guaranteeing zero leakage during the inference phase.

D Experimental Analysis

D.1 RTO Comparison under Baselines

To evaluate the generalizability and effectiveness of the RTO, we compare the Tail-CR@95 of the FTIF framework against several state-of-the-art baselines, including AutoXPCR, MetaEFormer, SDE, PatchTST, and iTransformer. Each model is evaluated under both the standard MSE objective and the proposed RTO.

As illustrated in Fig. 10, the integration of RTO loss yields a universal performance gain in Tail-CR@95 across all evaluated architectures, consistently outperforming the MSE baseline. While models trained with MSE exhibit Tail-CR@95 values between 0.2402 and 0.7012, the adoption of RTO elevates these metrics to a higher bracket of 0.2754 to 0.7412. Notably, our proposed FTIF achieves the peak Tail-CR@95 of 0.7412, significantly surpassing other RTO-enhanced models such as AutoXPCR (0.6230) and iTransformer (0.3867). This consistent upward trend underscores the robust capability of RTO loss to enhance tail-event capture across diverse model structures.

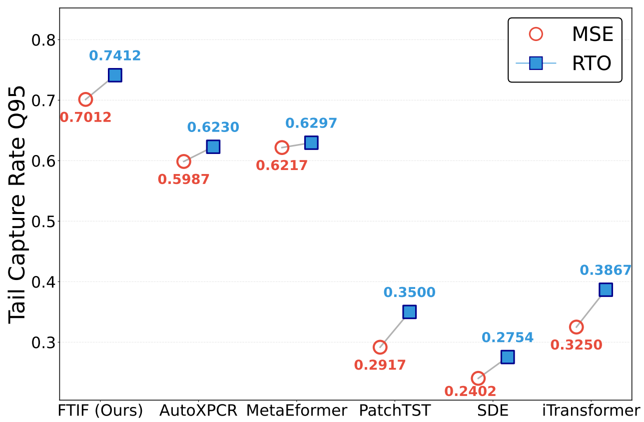


Figure 10: Comparative Tail-CR@95 for FTIF and baselines under MSE and RTO.

E Related Work

E.1 Decomposition-driven Time Series Representation

Mainstream research in temporal forecasting has shifted significantly toward interpretable deep decomposition. Autoformer [31] and FEDformer [37] established the classical architecture of embedded decomposition and frequency-domain truncation. They reduce forecasting difficulty by stripping robust trend components from high-noise data. Following this trajectory, DLinear [34] and TiDE [5] further demonstrated that simple explicit linear mappings suffice to outperform complex Transformer baselines, while TimeMixer [27] refined this representational granularity through a multi-scale mixing mechanism. Concurrently, Chronos [2] introduced patch-based tokenization strategies to enhance local semantic representation by reducing sequence redundancy.

However, such decomposition mechanisms based on smoothing filters or low-rank frequency approximation induce a severe Dominant Representation Bias in cloud workload scenarios. Massive normal load gradients mask sparse long-tail risk signals during back-propagation. This causes models to degenerate essentially into low-pass filters that misinterpret high-frequency micro-bursts (critical precursors to failure) as noise and filter them out.

E.2 Interaction Modeling based on Structural Priors

In hyperscale cloud environments, precise reconstruction of non-linear resource coupling is key to overcoming SLA assurance bottlenecks. Current research primarily evolves along two paths. The first is the Channel Independence (CI) strategy represented by PatchTST [17], which maximizes generalization robustness by completely blocking variable interactions. The second utilizes full attention mechanisms, such as iTransformer [15] and Crossformer [36], to explicitly model global dependencies. From the perspective of algebraic structural priors, methods like TUCKET [23] attempt to extract latent factors via tensor decomposition, while FBP [13] captures second-order interactions using bilinear pooling.

Nevertheless, the CI strategy ignores the physical coupling caused by resource fragmentation and fails to perceive the resonance of system states. Conversely, full attention mechanisms are computationally expensive for high-dimensional sparse metrics and introduce significant irrelevant noise. More critically, the Additive Inductive Bias inherent in traditional models lacks the algebraic capacity required to bridge the joint distribution of multi-dimensional tails. It struggles to capture high-order multiplicative synergies between variables. Consequently, these models fail to reconstruct the non-linear boundaries of multi-dimensional risks in extreme congestion scenarios.

E.3 Adaptive Forecasting in Non-Stationary

Addressing the dynamic heterogeneity of cloud workloads, researchers have developed various adaptive techniques [32]. DynEformer [11] leverages global receptive fields within a Transformer-based architecture to capture long-range temporal dependencies in non-stationary streams. To further handle evolving system dynamics, MetaEFormer [12] introduces meta-learning components to achieve rapid parameter adaptation. Meanwhile, TimeFilter [10] and OrgLinear [6] attempt to handle non-stationarity through structural decomposition, explicitly separating trend and seasonal components to mitigate distribution shifts.

Despite these advances, existing adaptive mechanisms are generally constrained by an optimization landscape dominated by MSE. This objective function allows gradient flows to be dominated by high-probability normal samples and prevents an acute response to irreversible rigidity in the state dimension. Most models can only adapt to first-moment drifts (mean shifts) in the distribution, but lack the ability to detect extreme-value characteristics that define SLA violation risks. The framework proposed in this paper aims to overcome this limitation by establishing a long-tail computation path independent of the normal distribution to precisely capture critical risk signals.